Reconstruction of the *Escherichia coli* LEE pathogenicity island network

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Abstract

Enteropathogenic *Escherichia coli* (EPEC) and enterohemorrhagic *Escherichia coli* (EHEC) have the locus of enterocyte effacement (LEE) pathogenic island, which contains the virulence genes that confer the ability to produce the Attaching and Effacing (A/E) lesion in humans causing acute diarrhea. Despite medical importance, there are few efforts to build the regulatory network of the genes related to this island. It has not been integrated all the information due to the complexity of global and local regulation. Here, we show a semi-automatic curation workflow to obtain the LEE pathogenicity island regulatory network from scientific articles through by extracting regulatory interactions and their properties. The regulatory network contains the genes belonging to the LEE island and some non-encoded genes regulated by LEE regulators in EPEC and EHEC strains. Our network comprises 239 regulatory interactions obtained from 250 papers, including 43 transcription factors and 84 target genes.

Introduction

Regulatory networks allow visualizing how biological systems are regulated, and they are being applied in several fields of knowledge such as medicine and biology (1). Emergent properties, i.e. properties that cannot be observed analyzing the behavior of each node alone, is one of the useful characteristics of the networks or feedback loops cha characterized to form circuits of positive or negative regulation which affect the synthesis rate of some components of the network (2). Because of this, our primary goal is to integrate all known information using assisted curation to have a global picture of how LEE island is being regulated. The interactions from this network are classified as either direct or indirect relationships and as on the experimental methods used to obtain them. The network integration allowed us to find new feedback loops that have not been previously integrated.

Funding

CONACYT PhD scolarship CONACYT to Sara B. Martínez-Luna

Acknowledgments

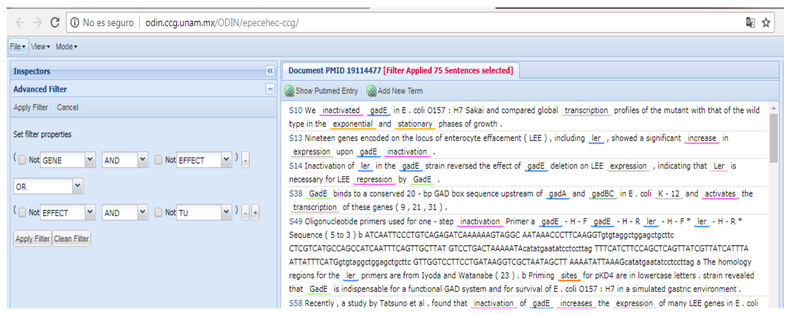
To Fabio Rinaldi

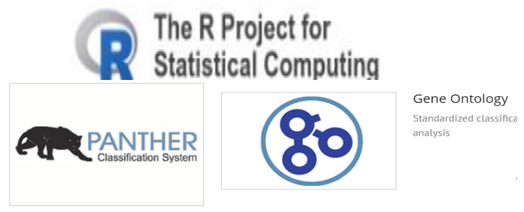
References

1. Emmert-streib, F., Dehmer, M., Haibe-kains, B., et al. (2014) Gene regulatory networks and their applications : understanding biological and medical problems in terms of networks, 2, 1–7.
2. Thomas, R. and D. R. Biological Feedback; 2003.

Methods

Set of 250 articles





Results

Transcriptional regulation

TF name	Gene/ Operon name	Effect	Growth conditions	Methods	References
GrlR-A (LEE7)	*ler* (LEE1)	Activation	Static cultures, shaken cultures, DMEM	Transcriptional fusion, EMSA	16291665, 14988506 20622062
Hfq	*grlR-A (LEE7)*	Repression	Exponential phase, LB	qRT-PCR, Northern blot and Microarray analyses	19570135

Table 1. Examples of Regulatory Interactions, their growth conditions, methods and references.

Another types of levels of regulation

- Post-transcriptional regulation (Fur regulates *ler* (LEE1) and pchABC) [25006810].
- Regulation by Proteases (ClpXP regulates *grlR-A* (LEE7), *rpoS* and *nlpD*) [20494990,19581394].
- Regulation by small RNAs (GlmY-repression LEE5 and LEE4) [27709103].

Table 2. In addition to the extraction of RIs OntoGene we could extract additional information such as other levels of regulation for each RI.

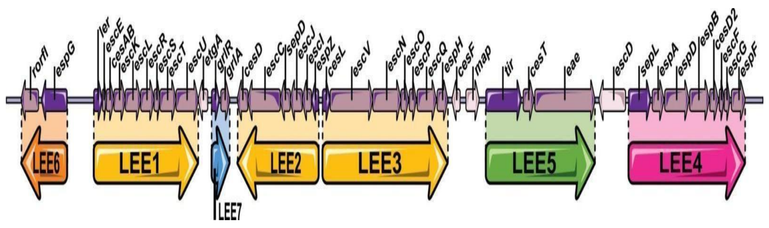


Figure 1. The LEE island contains 41 genes organized in seven operons (LEE1-LEE7).

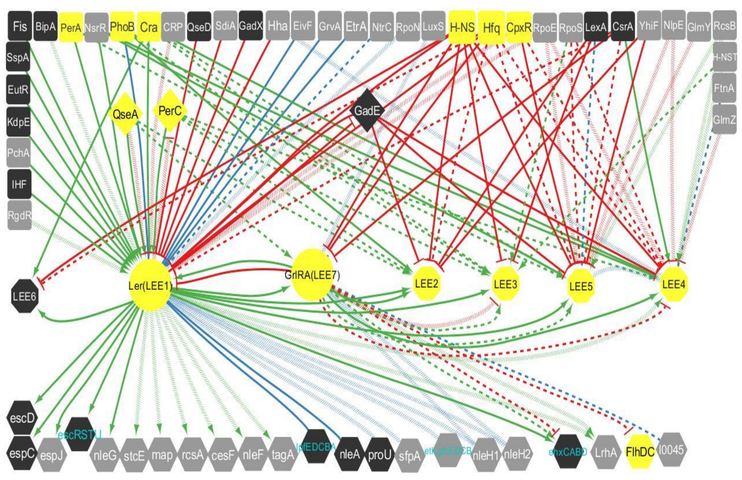


Figure 2. LEE enteropathogenic regulatory network. This figure represents the connections of the pathogenicity island LEE1 to LEE7

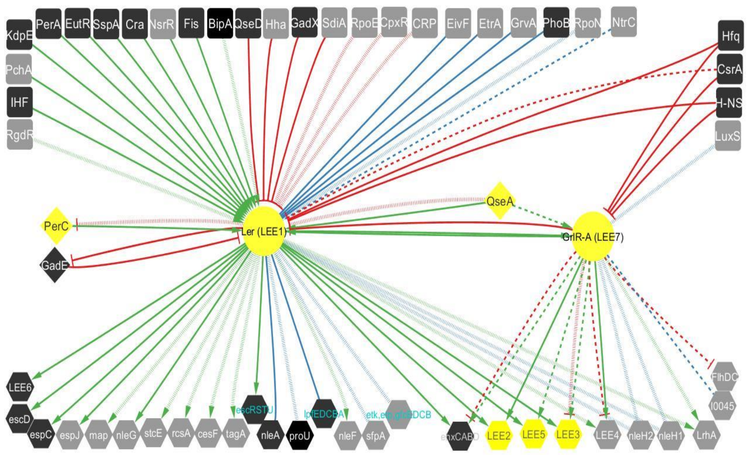


Figure 3. In this network composed of the two nodes with the highest degree of global connectivity (hub nodes) are: Ler with Ke = 32 and Ks = 32, the operon GrlR-A with Ke = 7 and Ks = 12, where Ke represents the number of incoming connections and Ks represents the number of outgoing connections

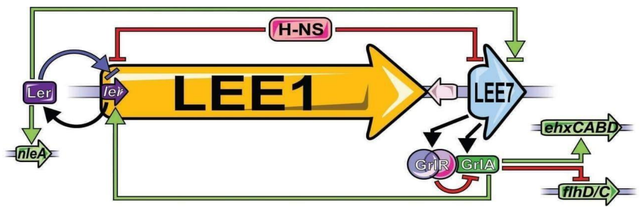


Figure 4. Feedback regulation system of the pathogenicity island LEE in EPEC and EHEC.

Conclusions

Assisted curation through ODIN clearly shows to be an efficient tool to curate data in EPEC and EHEC. Our network contains 85 regulatory interactions of which the representative and dual nodes are Ler (LEE1) and GrlR-A (LEE7) the master regulators of the pathogenicity island. A Gene Ontology analysis shows three of the main mechanisms used by EPEC and EHEC to develop pathogenesis in the intestine. Input and output connectivities (Ke, Ks) were calculated for each node of the network. Ke was defined as the number of the input edges of the node.